ML05 Lists, dictionaries and loops

Week 2, Python you need | Session 5 of 24 | Machine Learning

Today's question. How do I repeat one calculation for every item in a list?

Objective. By the end: you can read a list position and a dictionary key, and trace a loop that adds up values.

> **WHERE THIS IS USED**
> A football league table is a list of teams with points, and a program loops over it to add up every match. Your mobile money statement works the same way. What would you loop over in your favourite sport?

Before this session: ML04 Decisions and a hand-written flower rule. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
list	Several values kept in order inside square brackets.	A queue of customers.
index	The position of an item in a list, counting from 0.	The house number of a plot.
dictionary	Pairs of a name and a value inside curly brackets.	A phone book of names and numbers.
key	The name used to look up a value in a dictionary.	The name you look up in a phone book.
loop	A for statement that repeats indented lines for every item.	Giving each pupil in a row a book.
item	One value taken from a list during a loop.	The pupil currently getting a book.

Block A. Lists and dictionaries

A tuck shop keeps three prices in a list and one product in a dictionary.

Predict. Predict what prints. Which number is prices[0]? Write your guess on paper first.

Your guess, written before you run it

```
prices = [1000, 2000, 3000]
product = {"name": "Juice", "price": 2000}
first = prices[0]
juice_price = product["price"]
print(first)
print(juice_price)
print(len(prices))
```

Run it, then compare. The cell prints 1000, 2000 and 3. Square brackets after a list give the item at that index, and the first index is 0. Square brackets after a dictionary give the value for that key. len counts the items in a list.

Investigate.

1. Why is prices[0] the first price?
2. What happens if you ask for prices[3]?

Modify. Print the last price using its index.

Block B. A loop that adds

Run the loop and follow how total grows after each pass.

Predict. Predict the four numbers printed. Write your guess on paper first.

Your guess, written before you run it

```
prices = [1000, 2000, 3000]
total = 0
for price in prices:
    print(price * 2)
    total = total + price
print(total)
```

Run it, then compare. The cell prints 2000, 4000, 6000 and then 6000. for price in prices takes one item at a time and runs the indented lines for it. total starts at 0 and grows by the item each pass: 1000, then 3000, then 6000.

Investigate.

1. What is total after the second pass?
2. What goes wrong if total = 0 is moved inside the loop?

Modify. Change the list to [1500, 2500] and run the loop again.

Make

Make the list [2, 4, 6] and print each value plus 1 with a loop. Make a dictionary with the keys name and stock and print the stock. Then loop over the eight petal lengths 1.4, 1.4, 4.7, 4.5, 4.9, 6.0, 5.1, 5.9 and count how many are below 2.5.

Save your work as ML05. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
IndexError: list index out of range	You asked for a position the list does not have.	A list of 3 items has positions 0, 1 and 2.
KeyError: 'price'	The dictionary has no key spelled like that.	Check the exact spelling and capital letters of the key.

Exit ticket

A loop repeats the same lines for every item, and a total that starts at 0 grows each pass.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.